

Curriculum vitae

Personal Information

FIRST NAME / SURNAME	Dr. Tatiana Morozova
DATE OF BIRTH	24.12.1992
NATIONALITY	Russian
CIVIL STATUS	married
EMAIL	tatiana.morozova@ens-lyon.fr

Education

DATES	Mar. 2016 - Oct. 2019
QUALIFICATION PROGRAM	Ph.D. in Physics. Thesis on “Polymers and Polymer Colloids in Solvent Mixtures”, under the supervision of Dr. A. Nikoubashman
INSTITUTION	Johannes Gutenberg University, Mainz, Germany

DATES	Sept. 2010 - Jan. 2016
QUALIFICATION AWARDED	Specialist in Physics (higher education program in Condensed Matter Physics equaled a Master's Degree). Thesis on “Computer Simulation of Mechanical Properties of Polyelectrolyte Micro- and Nanocapsules”, under the supervision of Prof. Dr. O. I. Vinogradova
INSTITUTION	Lomonosov Moscow State University, Moscow, Russia

DATES	Sept. 1999 - Jun. 2010
QUALIFICATION AWARDED	High school Diploma
INSTITUTION	Gymnasium № 1519, Moscow, Russia

Work Experience

DATES	Dec. 2024- <i>present</i>
POSITION	Permanent researcher (Chargée de Recherche)
INSTITUTION	CNRS, Physics Laboratory at ENS Lyon, Lyon, France

DATES	Dec. 2023 - Nov. 2024
POSITION	Postdoctoral researcher
INSTITUTION	CNRS, Institut Pasteur, Paris, France

DATES	Oct. 2020 - Nov. 2023
POSITION	Postdoctoral researcher
INSTITUTION	Institut Max von Laue - Paul Langevin (ILL), Grenoble, France

DATES	Nov. 2019 - Sept. 2020
POSITION	Postdoctoral researcher
INSTITUTION	LIPhy, CNRS, University Grenoble Alpes, Grenoble, France

DATES	Mar. 2016 - Oct. 2019
POSITION	Ph.D. student
INSTITUTION	Johannes Gutenberg University, Mainz, Germany

Teaching Experience

DATES	March 2026
POSITION	Teaching assistant for the course “Molecular dynamics” (M1)
INSTITUTION	ENS Lyon, Lyon, France
DATES	Winter semester 2017/2018 & Summer semester 2018
POSITION	Teaching assistant for the course “Mathematics for Physicists”,
INSTITUTION	Johannes Gutenberg University, Mainz, Germany
DATES	Summer semester 2016
POSITION	Teaching assistant for the course “Advanced Statistical Physics”,
INSTITUTION	Johannes Gutenberg University, Mainz, Germany

Fundings Raised

DATES	2025-2029
NAME	Jeunes Chercheuses et Jeunes Chercheurs (JCJC) starting grant for young scientists based in France (ANR): “In silico: Disordered Proteins in Cells and Bio-Inspired Materials” (€ 241,000).
DATES	2025
NAME	CNRS DIALOG : computational equipment (€ 46,000)
DATES	2024
NAME	Marie Skłodowska-Curie Action (MSCA) European Postdoctoral Fellowships, project: “In silico modeling of multi-domain proteins in biological condensates” (231,000€). <i>Declined</i> for the permanent research position at CNRS.
DATES	2022 - 2025
NAME	Franco(ANR)-German(DFG) project: “Dynamics, kinetics and assembly of model intrinsically disordered proteins from a polymer physics perspective” in collaboration with F. Schreiber, M. Oettel, J.-L. Barrat, O. Matsarskaia, T. Seydel (650,000€).
DATES	2020 - 2026
NAME	Computational time at the French national cluster facilities IDRIS (CNRS) (300,000 GPU hours granted ~ 110,000€)
DATES	2017
NAME	Travel grant from the IRTG/TRR146 graduate school (3,000 €)

Research Visits

DATES	July 2022
PURPOSE	Collaborative visit to Dr. A. Deblais
INSTITUTION	Institute of Physics, University of Amsterdam, Netherlands

DATES	June - July 2017
PURPOSE	Experimental project on fabrication of polymer colloids in the research groups of A.Z. Panagiotopoulos, R.K. Prud'homme, R.D. Priestley
INSTITUTION	Department of Chemical and Biological Engineering, Princeton University, USA

Skills and Competences

LANGUAGES SPOKEN	Russian (native), English (fluent), French (B1), German (intermediate)
PROGRAMMING	C/C++, Python
SIMULATION SOFTWARE	HOOMD-Blue, GROMACS, PyEmma, PLUMED, AlphaFold

Organization of Scientific Events

DATES	Jul. 2021 - Oct. 2023
ROLE	College secretary of the Theory Group - organization of seminars & group activities, research visits, and workshops at ILL
DATES	Feb. - Oct. 2020
ROLE	Organizer of group seminars of the Statistical Physics and Modeling team (LIPhy, UGA/CNRS)
DATES	Oct. 2018 - Oct. 2019
ROLE	Ph.D. student representative - organization of seminars
DATES	May 2019
ROLE	Organization of the graduate students (CRC TRR146) retreat (5 days, Amsterdam, Netherlands)

Publications

- (1) **T.I. Morozova** and A. Nikoubashman: Coil-globule collapse of polystyrene chains in tetrahydrofuran-water mixtures, *J. Phys. Chem. B* 122, 2130 (2018)
- (2) **T.I. Morozova**, V.E. Lee, A.Z. Panagiotopoulos, R.K. Prud'homme, R.D. Priestley, and A. Nikoubashman: On the stability of polymeric nanoparticles fabricated through rapid solvent mixing, *Langmuir* 35, 709 (2019)
- (3) **T.I. Morozova** and A. Nikoubashman: Surface activity of soft polymer colloids, *Langmuir* 35, 16907 (2019)
- (4) **T.I. Morozova**, V.E. Lee, N. Bizmark, S.S. Datta, R.K. Prud'homme, A. Nikoubashman, R.D. Priestley: In silico design enables the rapid production of surface-active colloidal amphiphiles, *ACS Cent. Sci*, 6, 166 (2020)
- (5) **T.I. Morozova**, N.A. Garcia, J.-L. Barrat, G.S. Luengo, F.

Leonforte: Adsorption and desorption of polymers on bio-inspired chemically structured substrates, *ACS Appl. Mater. Interfaces*, 13, 25, 30086–30097 (2021)

(6) **T.I. Morozova**, N.A. Garcia, J.-L. Barrat: Temperature dependence of thermodynamic, dynamical, and dielectric properties of water models, *J. Chem. Phys.*, 156 (12), 126101 (2022)

(7) M.R. Bittermann, **T.I. Morozova**, S.F. Velandia, A. Deblais, S. Woutersen, D. Bonn: Surface-mediated molecular transport in polydisperse oil-in-water emulsions, *Langmuir*, 39, 12, 4207-4215 (2023)

(8) **T.I. Morozova**, N.A. Garcia, O. Matsarskaia, F. Roosen-Runge, J.-L. Barrat: Structural and Dynamical Properties of Elastin-Like Peptides near Lower Critical Solution Temperature, *Biomacromolecules* 24, 4, 1912-1923 (2023); **top twelve publications** by researchers at the ILL (*Highlights* 2023)

(9) I. Adroher-Benítez, **T.I. Morozova**, N.A. Garcia, J.-L. Barrat, G.S. Luengo, F. Leonforte: Modeling the adsorption of copolymers on heterogeneous surfaces and the stability of polymer coatings for cosmetic applications, *Macromolecules*, 56, 24, 10285-10295 (2023)

(10) **T.I. Morozova**, N.A. Garcia, J.-L. Barrat: Sequence length controls coil-to-globule transition in elastin-like polypeptides, *J. Phys. Chem. Lett.*, 15, 10757-10762 (2024)

(11) S. Chakraborty, **T.I. Morozova**, J.-L. Barrat: Intrinsically disordered proteins can behave as different polymers across their conformational ensemble, *J. Phys. Chem. B*, 129, 2359-2369 (2025)

(12) V. Schnapka, **T. I. Morozova**, S. Sen, and M. Bonomi: Atomic resolution ensembles of intrinsically disordered and multi-domain proteins with AlphaFold, *Nat. Commun.*, 17, 2399 (2026)

(13) L. M. Miñarro, S. Chakraborty, C. Beck, A. C. Grundel, I. Mosca, F. Roosen-Runge, **T. I. Morozova**, J.-L. Barrat, F. Schreiber, and T. Seydel: Non-Fickian diffusion within assemblies of the intrinsically disordered protein β -casein”, *PNAS*, 123, 11, e2532636123 (2026)

Preprints

(i) S. Sen, S. E. Hoff, **T. I. Morozova**, V. Schnapka, and M. Bonomi: Advancing in silico drug design with Bayesian refinement of AlphaFold models (2025)

Talks and Posters

Seminar: Laboratoire de Physique ENS de Lyon, Lyon, 2023: “Thermoresponsive polypeptides: insights from the polymer physics”

Seminar: Laboratoire de biochimie théorique, IBPC, Paris, 2023: “Phase behavior of thermo-responsive polypeptides”

Seminar: ICS Strasbourg, Strasbourg, 2022, *Structural and Dynamical Properties of Elastin-Like Peptides near Lower Critical Solution Temperature*

Seminar: Thematics School in Soft Condensed Matter (Univ, Grenoble Alpes) 2021, *Fabrication of Polymer Colloids through Rapid Solvent Exchange*

Seminar: University of Pennsylvania 2019, *Computer simulations of controlled fabrication of polymer colloids through the rapid solvent exchange*

Invited Talk: conference FisMat 2025 (Venice, Italy): *Disordered proteins: insights from polymer physics*

Contributed Talk: CECAM workshop: Frontiers of Coarse-Grained Models: From New Developments to Modeling Dynamics, Assemblies, and Macromolecular Machines, Lyon (France), 2024: *Thermo-responsive polypeptides: insights from polymer physics*

Contributed Talk: 28th International Conference on Statistical Physics (Statphys28), Tokyo (Japan), 2023: *Coil-to-globule transition of thermoresponsive polypeptides*

Contributed Talk: Spring meeting of the American Chemical Society (online) 2021, *Polymer adsorption on bio-inspired disordered substrate*

Contributed Talk: Spring meeting of the DPG, Regensburg (Germany) 2019, *Fabrication of polymeric Janus nanoparticles and their behavior at the liquid-liquid interfaces - a simulation study*

Contributed Talk: Spring meeting of the DPG Berlin (Germany) 2018, *Polymer self-assembly into nanoparticles through rapid solvent exchange in organic media*

Contributed Talk: 91st ACS Colloid&Surface Science Symposium, New York City (USA) 2017, *Atomistic simulations of the coil-globule transition of polystyrene chains in solution*

Contributed Talk: Spring meeting of the DPG, Dresden (Germany) 2017, *Controlled fabrication of nanoparticles through rapid solvent exchange*

Poster: CECAM workshop: Development of Coarse-Grained Models, Lyon (France) 2022, *Adsorption and desorption of polymers on bio-inspired chemically heterogeneous surfaces*

Poster: Les Houches-TSRC Protein Dynamics Workshop, Aussois (France) 2022, *Structural and dynamical properties of elastin-like peptides near lower critical solution temperature*

Poster: CECAM workshop: From disordered biomolecular complexes to biological coacervates, Zurich (Switzerland) 2022, *Structural and dynamical properties of elastin-like peptides near lower critical solution temperature*

Poster: 5th Edwards Symposium, Cambridge (UK/online) 2021, *Adsorption and desorption of polymers on bio-inspired chemically heterogeneous surfaces - poster prize*

Poster: 3rd Edwards Symposium, Cambridge (UK) 2018, *Fabrication of Janus nanoparticles from the self-assembly of homopolymers and amphiphiles through rapid precipitation*

Poster: CECAM workshop “Collective behavior of soft and active matter under confinement”, Mainz (Germany) 2018, *Fabrication of Janus nanoparticles from the self-assembly of homopolymers and amphiphiles through rapid precipitation*